

**Mrs White
AP Chemistry
Daily Agenda Slides**



**Unit 5:
Thermochemistry**

U5D1 November 10-11 (M/T) -

Learning Target: What is the difference between an endothermic and exothermic process?

Agenda:

- ~Review? Do you have questions before the test?
- ~Thermochemistry day 1!

Homework:

- ~Study Study Study!
- ~Textbook Probs due next time
- ~Test next time

U4 Test Day November 12-13 (W/Th) -

Learning Target: What do you know about EQUILIBRIUM?

Agenda:

~Unit 4 Exam

Homework:

~None!

U5D2 November 14-17 (F/M) - Mrs White Sub Day

Learning Target: How do we measure heat change?

Agenda:

- ~Watch the rest of the U5 Lecture 1 and fill in the notes through page 2
- ~Watch U5 Lecture 2 and fill in the notes through the end of 4- try the MCQ
- ~Watch U5 Lecture 3 and fill in the notes through the end of p 5
- ~Try the Heating Curve Calculations Worksheet (key is on the front table)
- ~Still want more? Look at the U5 Textbook Practice 1

Homework:

- ~Finish through Lecture video 3 before next time- **there will be an open note quiz**
- ~U5 Textbook Problem Set 1 due 12/2-3 @ 11:55
- ~U5 Test 12/15-16

U5D3 November 18-19 (T/W) -

Learning Target: What is the heat of a reaction?

Agenda:

~QUIZ! Over first 3 lectures

~Review Exam 4

~Reaction Enthalpy notes and practice

Homework:

~U5 Textbook Problem Set 1 due 12/2-3 @ 11:55

~U5 Test 12/15-16

U5D4 November 20-21 (Th/F) -

Learning Target: How can the heat of reaction be calculated from standard values?

Agenda:

~Take out your Chromebook at take this [survey](#)!

~WARM UP ON BIG WHITEBOARD Heating and Cooling Curve Calculations- 150.0 mL of ice at 0.0°C is heated until it becomes water at room temperature, 25.0°C . Draw a graph representing this process. Calculate the enthalpy for this process. The heat capacity of water is $4.184\text{ J/g}^{\circ}\text{C}$, and the heat of fusion of water is 6.04 kJ/mol .

~More heat of reaction calculations- Hess's Law and Heats of Formation equation

Homework:

~U5 Textbook Problem Set 1 due 12/2-3 @ 11:55

~U5 Textbook Problem Set 2 due 12/14-15 @ 11:55

~U5 Test 12/15-16

~Lab next time! Closed Toe Shoes

U5D5 December 1-2 (M/T) -

Learning Target: Can the heat of reaction be found through experimental methods?

Agenda:

- ~Warm up- Let's try that BIG MAMA equation again!
- ~Investigation: Is recycling worth the heat?
- ~Lab: Heats of Reaction

Homework:

- ~U4 test corrections by 12/5!
- ~Finish Lab (if needed)
- ~U5 Textbook Problem Set 1 due 12/2-3 @ 11:55
- ~Quiz! 12/5-8
- ~U5 Textbook Problem Set 2 due 12/14-15 @ 11:55
- ~U5 Test 12/15-16

Is Recycling Worth the Heat?

Background:

Do you recycle your aluminum cans? Let's investigate the energy associated with different ways to get aluminum. Aluminum metal can be recycled from scrap metal by melting the metal to evaporate the impurities. Aluminum can also be obtained by extracting it from its ore, Al_2O_3 .

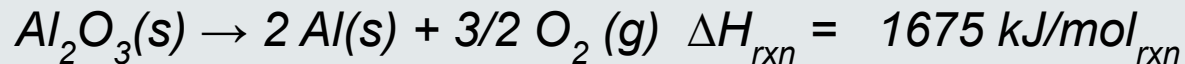
Question:

Which requires less energy, recycling 50.0g of existing Al or extracting 50.0 g Al from Al_2O_3 ?

Here are some hints.

Recycling method: To recycle 50.0 g of aluminum, assume its initial temperature is 25°C and needs to first be heated up to its melting point of $660.^\circ\text{C}$, then melted. The heat capacity of Aluminum is $0.890 \text{ J/g } ^\circ\text{C}$. The heat of fusion is 10.7 kJ/mol .

Extracting from aluminum ore: The equation for the process of extracting Al from Al_2O_3 is shown below.



U5D6 December 3-4 (W/Th) -

Learning Target: How else can heat of reaction be determined?

Agenda:

- ~Warm Up- Try the MCQ on pp. 10-11
- ~One more method to get MOLAR ENTHALPY: Bond Energies
- ~Let's get chaotic- Entropy

Homework:

- ~Quiz 12/5-8
- ~U5 Textbook Problem Set 2 due 12/14-15 @ 11:55 pm
- ~U5 Test 12/15-16

U5D7 December 5-8 (F-M)

Learning Target: What is entropy?

Agenda:

~Group Quiz

~Notes and Practice: Entropy and Gibbs Free Energy

Homework:

~U5 Textbook Problem Set 2 due 12/14-15 @ 11:55 pm

~U5 Test 12/15-16

U5D8 December 9-10 (T/W)

Learning Target: How is equilibrium related to free energy?

Agenda:

- ~Warm Up! Try #1-3 in the back page of your packet
- ~Notes and practice: Equilibrium and Gibbs Free Energy
- ~Review for exam

Homework:

- ~U5 Textbook Problem Set 2 due 12/14-15 @ 11:55 pm
- ~U5 Test 12/15-16

U5D9 December 11-12 (Th/F)

Learning Target: What have you learned about thermochemistry?

Agenda:

- ~Take out your chromebook and fill out this [SURVEY](#)!
- ~Check HAC- are there grades that need to be updated? Fill in the Grade Change Form. LAST CHANCE to submit work is Wednesday 12/17!
- ~Review Challenge!

Homework:

- ~Textbook Prob Set #2
- ~Unit 5 exam next time

U5D10 December 15-16 (M/T)

Learning Target: What have you learned about thermochemistry?

Agenda:

~Unit 5 exam

Homework:

None!